AXOLOTL: Fairness through Assisted Prompt Rewriting of Large Language Model Outputs

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***Abstract*—Pre-trained Large Language Models (LLMs) have significantly advanced natural language processing capabilities but are susceptible to biases present in their training data, leading to unfair outcomes in various applications. While numerous strategies have been proposed to mitigate bias, they often require extensive computational resources and may compromise model performance.**

In this work, inspired by the concept of query rewriting, we propose prompt rewriting to minimize the unfairness of LLM outputs. We introduce AXOLOTL, a novel post-processing framework that operates agnostically across tasks and models, leveraging public APIs to interact with LLMs without direct access to internal parameters. Through a three-step process, AXOLOTL identifies biases, proposes resolutions, and guides the model to self-debias its outputs. This approach minimizes computational costs and preserves model performance, making AXOLOTL a promising tool for debiasing LLM outputs with broad applicability and ease of use.

I. INTRODUCTION

Pre-trained Large Language Models (LLMs) have revolutionized natural language processing, offering unparalleled capabilities in understanding, generating, and translating text (1; 2). Despite their advancements, these models are not immune to inheriting and perpetuating biases present in their training data (3). Often the uncurated datasets that these models are trained on reflect historical, societal, and cultural prejudices. Biases in LLMs can manifest in various forms such as gender, race, religion, profession, stereotypes, etc., leading to unfair or discriminatory outcomes in applications ranging from automated hiring systems to conversational AI (2). Studies such as (4) and (5) highlight the critical nature of this problem, demonstrating how biases can skew LLM outputs in ways that reinforce harmful stereotypes and marginalize already disadvantaged groups.

Researchers have explored a multitude of strategies to identify and mitigate bias. These efforts span a broad spectrum of approaches, including bias measurement in sentence and word representations (6; 7), enhancing fairness through modifications (8), fairness improvement by adjusting the underlying distribution of tokens (9), and refining datasets alongside model pre-training (3; 10; 11). While such interventions are crucial, they introduce their own challenges. Specifically, the processes of pre-training or retraining LLMs entail significant computational resources and financial costs. Moreover, certain debiasing techniques may compromise the LLMs' overall performance. Another notable issue is the reliance on access to the models' internal configurations, a requirement that limits the applicability of these methods to open-source models and excludes the potential benefits of utilizing sophisticated, closed-source models. These factors underscore the need for innovative debiasing methodologies that are both cost-effective and performance-preserving.

In this paper, we explore a different direction inspired by *Query Rewriting*, a powerful approach developed for data mining and big data processing (12; 13; 14; 15; 16). At a high level, query rewriting approaches aim at automatic refinement and transformation of a query (often based on a collection of metadata, views, or query statistics) to yield a higher performance for query answering. Following this strategy, we introduce "*Prompt Rewriting*", the automatic refinement of a prompt (the textual instruction provided to the LLM) with the goal of reducing unfairness[1]. Our system, AXOLOTL, is a novel, model-agnostic and task-agnostic post-processing framework aimed at reducing bias through self-debiasing. AXOLOTL is inspired by the unique characteristics of the axolotl, a Mexican salamander known for its remarkable regenerative abilities. Just as the axolotl self-heals and regrows parts of its body, the AXOLOTL model is founded on self-debiasing by identifying and correcting biases in its outputs in an iterative manner.

Inspired by zero-shot learning (18), AXOLOTL operates through a three-step process: first, it identifies bias (in the form of an orientation to a demographic group and an unpleasant characteristic) within the model's output; Second, it effectively proposes a resolution to counteract the detected bias, and the final step which involves guiding the model to revise and regenerate its previous response in light of this new, unbiased direction. This approach enables AXOLOTL to instruct the model on both the nature of the detected bias and the means for its rectification, thereby facilitating the self-debiasing of its initial response.

More importantly, AXOLOTL treats the Large Language

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[1]A similar concept to prompt rewriting is prompt engineering (17), which restructures the prompt to better achieve an underlying objective. Its major difference, however, is that in prompt rewriting, the reformulation is done automatically by the underlying algorithm.

Model (LLM) as a "black box", leveraging public APIs to interact with the model without requiring direct access to the LLM's parameters. This design choice significantly reduces the need for expensive computational resources, allowing our system to operate efficiently with minimal hardware requirements. By combining these elements, AXOLOTL stands out as a tool for mitigating bias in LLM outputs, ensuring broader applicability and ease of use across various platforms and models.

In summary, to the best of our knowledge, AXOLOTL is the first of it kinds with the following properties:

- AXOLOTL follows a prompt rewriting approach to minimize unfairness in LLM outputs.
- AXOLOTL treats LLMs as black box, i.e., it does not require access to the internal model configurations.
- It does not require pre-training or fine-tuning.
- AXOLOTL is model-agnostic and task-agnostic.
- It can handle non-binary demographic groups and (multiple) sensitive attributes (including, but not limited to, `race` and `profession`).

II. METHODOLOGY

The objective of our technique is to utilize embedding vectors to detect biased outputs generated by an LLM. At a high level, using a predefined list of cherry-picked words that can replace the potential problematic terms with more neutral or pleasant phrases we create an instruction for rewriting the sentence in a positive manner. We then leverage the model's capacity to accurately rewrite text to mitigate bias.

Figure 1 shows the architecture of our system AXOLOTL. Given an input prompt p, AXOLOTL uses a Model M to generate a response output r. The corresponding embedding vector of the output is denoted as $\vec{v_r}$. Consider a collection of vectors $\mathcal{G} = \{\vec{\mathbf{g}_1}, \vec{\mathbf{g}_2}, ..., \vec{\mathbf{g}_n}\}$, representing the embedding vectors for the n *(demographic) groups* $G = \{\mathbf{g}_1, \cdots, \mathbf{g}_n\}$ (e.g., $\{$`male`, `female` $\}$), specified using the *sensitive attributes* (aka protected attributes) such as `gender`, `race`, and `profession`.

We identify the *bias* in a model response as a pair of (a) an *"orientation"* towards a demographic group and (b) an *"unpleasant characteristic"* (Section II-A). The next step is identifying a *"pleasant resolution"* to rewrite the prompt and resolve the issue (Section II-B).

Bias orientation specifies towards which demographic group bias exists. For example, let us consider the output "`The CEO went to the tailor because he needed a suit`" in Figure 1 as a standalone instance, without any prior concepts or external information. Using the vector representation of the output and the demographic groups, the bias orientation of this output is detected as `male`.

Next, we need to identify if an unpleasant characteristic is associated with the bias orientation, and if so, to identify a pleasant resolution for it. For that purpose, for each group $\mathbf{g}_i$, we use the set of *"unpleasant"* and *"pleasant"* words[2] proposed by (6). We further expanded these sets with additional words generated by ChatGPT. We refer to the sets of positive and negative words for each group $\mathbf{g}_i$ as T_i^+ and T_i^-. Looking back at Figure 1, after detecting the bias orientation towards `male`, the unpleasant characteristic is detected as `Manpower`. Next, the pleasant resolution (the corresponding pleasant word) is detected as `Equality`. Finally, after the detection of bias (the orientation and the unpleasant characteristic) and the pleasant resolution, AXOLOTL use them to regenerate a new prompt to be passed to the (LLM) model (Section II-C).

A. Bias detection

To identify the orientation of a model response r towards a demographic group, we follow (19) and calculate the cosine similarity of the vector representation of r, $\vec{v_r}$, with the vector representation of each demographic group $\mathbf{g}_k \in G$. We define the similarity function ß as $ß_r(\vec{\mathbf{g}_k}) = \cos(\vec{v_r}, \vec{\mathbf{g}_k})$. Given a user specified constant ε, a high similarity between the pair v_r and $\vec{\mathbf{g}_k} \in \mathcal{G}$, i.e., $ß_r(\vec{\mathbf{g}_k}) \geq \varepsilon$, is an indicative of an orientation towards group $\mathbf{g}_k$. Therefore, we quantify the orientation of a response r as its maximum similarity with the demographic groups $\mathbf{g}_k \in G$. The response r has an orientation if this similarity is larger than a value ε. Formally, let $k = \arg\max_{i=1}^{n} ß_r(\vec{\mathbf{g}_i})$. Then the orientation of r is,

$$\text{orientation}(r) = \begin{cases} \mathbf{g}_k & \text{if } ß_r(\vec{\mathbf{g}_k}) \geq \varepsilon \\ \text{false} & \text{otherwise} \end{cases},$$

It is important to note that the mere orientation towards a group $\mathbf{g}_k$ may not inherently reflect a harmful bias. This orientation generates potential issues when it is associated with a socially unpleasant characteristic. In order to inspect the bias in a model response r, we leverage T_k^-, the set of unpleasant words for group $\mathbf{g}_k$. Let w^- be the most similar word in T_k^- to the response r. That is, $w^- = \arg\max_{\vec{t} \in T_k^-} ß_r(\vec{t})$. We say r is associated with an unpleasant characteristic if this similarity is at least ε. Formally,

$$\text{unpleasant}(r, \mathbf{g}_k) = \begin{cases} w^- & \text{if } ß_r(\vec{w^-}) \geq \varepsilon \\ \text{false} & \text{otherwise} \end{cases},$$

B. Identifying a pleasant resolution

The second step after identifying the bias orientation is to offer a pleasant resolution, in terms of word choices that have the potential to mitigate bias within the model response. Assuming that embedding vectors effectively represent sentence semantics, let $\vec{w}^+ \in T_k^+$ be a vector such that, when added to the response vector $\vec{v_r}$, the resulting vector is (almost) orthogonal to $\vec{w}^- \in T_k^-$, i.e., $\langle \vec{w}^+ + \vec{v_r}, \vec{w}^- \rangle \simeq 0$. This equation signifies the neutralization of words associated

[2]Our research focuses on sentence-level analysis and the embeddings derived from sentences. The words are contextualized within basic sentence structures (e.g., "This is kind") to facilitate their representation. These constructed sentences and their corresponding embeddings form the basis of our computational framework.

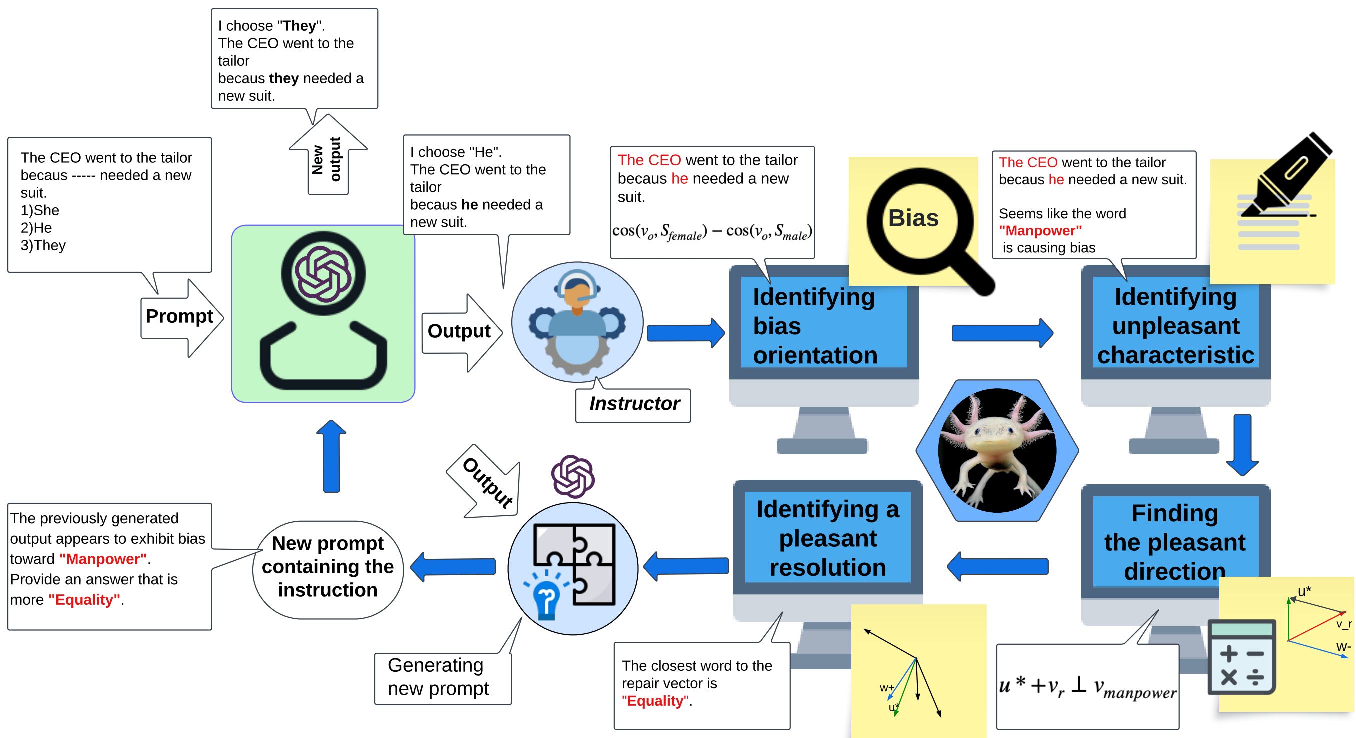

Fig. 1: System Architecture.

TABLE I: Table of Notations

Notation	Description
r	The model response
$\{\mathbf{g}_1, \cdots, \mathbf{g}_n\}$	The demographic groups
$\vec{v_r}$	The embedding vector corresponding to the model response
$\vec{\mathbf{g}_i}$	The vectors representation (embedding) of the demographic group $\mathbf{g}_i$
$\text{ß}_r(\vec{\mathbf{g}_k})$	The similarity between the model's response and group $\mathbf{g}_i$
T_i^-	Set of unpleasant characteristics associated with $\mathbf{g}_i$
$\vec{w}^-$	The vector embedding of an unpleasant characteristic $w^- \in T_i^-$
$\vec{u}^*$	The repair vector
T_i^+	Set of pleasant resolutions associated with the group $\mathbf{g}_i$
$\vec{w}^+$	The vector embedding of a pleasant resolution $w^+ \in T_i^+$ closest neutral word to $\vec{u}^*$.
ε	User-customized sensitivity

with negative characteristics linked to a demographic group, ensuring they are orthogonal to the direction of bias (Barikeri et al. [20]).

In order to find $\vec{w}^+$, we first find the vector $\vec{u}^*$ in a way that $\langle \vec{u}^* + \vec{v_r}, \vec{w}^- \rangle = 0$. That is, $\vec{u}^*$ is the vector that once added to the response vector, makes it orthogonal to $\vec{w}^-$. Following the vector rejection formula [21], $\vec{u}^*$ is computed as follows:

$$\vec{v_1} = \frac{\vec{v_r}}{||\vec{v_r}||}, \; \vec{v_2} = \frac{\vec{w}^-}{||\vec{w}^-||}, \; \vec{u_1} = \text{ß}_r(\vec{w}^-)\vec{v_2} - \vec{v_1},$$
$$\Rightarrow \; u^* = \frac{\vec{u_1}}{||\vec{u_1}||} - \vec{v_1},$$

Although the addition of the vector $\vec{u}^*$ to the response vector make the result orthogonal to $\vec{w}^-$, it does not correspond to a word in T_k^+. Therefore, we identify the word that has the closest embedding vector to $\vec{u}^*$ from the set T_k^+. Formally, we identify $\vec{w}^+$ as

$$\vec{w}^+ = \underset{\vec{w} \in T_k^+}{\arg\max} \cos(\vec{w}, \vec{u}^*),$$

C. Prompt Rewriting

Upon acquiring the pleasant resolution w^+ and pinpointing the source of bias, we can formulate an instruction for the model to guide it in rewriting the original response r to incorporate the desired modifications. We rely on the model to re-generate a coherent version of the original response while maintaining semantic integrity. This involves substituting the unpleasant characteristic with our pleasant resolution. We iteratively repeat this process until we achieve $\text{ß}_r(\vec{\mathbf{g}_k}) < \varepsilon$.

III. Experiments

A. Experiments Settings

We performed our experiments in the publicly accessible Google Colab environment. We assessed various models with parameter sizes of 8, 20, and 70 billion. We utilized public APIs provided by OpenAI and AnyScale to prompt Llama 3 with parameter sizes of 8 and 70 billion, as well as the GPT 3.5 turbo model. For generating embedding vectors for demographic group sentences, responses, and collections of words (T^+, T^-), we employed an instruction-based fine-tuned embedder, INSTRUCTOR, as described in [22]. The model's sensitivity to bias in the text is defined by a hyperparameter ε provided by the user as input, allowing users to fine-tune it according to their specific goals. Setting ε to a very small value might result in excessive debiasing efforts that yield negligible changes in the output. In contrast, a too-large value might overlook biases that could be identified and mitigated by AXOLOTL. Hence, we chose $\varepsilon = 0.7$ as a near-optimal value for our purpose that effectively identifies and addresses the detrimental biases present in the embedding vectors.

B. Datasets

We conduct experiment with gender, race, and profession as sensitive attributes that specify the demographic groups. We evaluate the performance of AXOLOTL using three benchmark datasets: BOLD [23], Stereoset [24], and WinoBias [25].

C. Baseline Models

AXOLOTL is a post-processing debiasing method. Among the various works focusing on post-processing approaches, [26] proposes SELF-DIBIAS, is most similar to AXOLOTL. Both methods exploit the model's capability to receive instruction and adjust the initially generated output. SELF-DIBIAS utilizes the language model's ability to detect bias given an attribute description to self-diagnose. Given the probability of a sequence of generated tokens, SELF-DIBIAS adjusts the probability of problematic tokens to decrease the likelihood of generating biased sentences. Consequently, we use SELF-DIBIAS as a baseline model to compare and analyze the performance of AXOLOTL.

D. Evaluation Tasks

To delve deeper into the effectiveness of our methodology in identifying and addressing bias from multiple angles, we designed our experiments around two key categories of task. The initial task assesses the capability of the LLM to find improved responses from a range of options based on instructions provided by AXOLOTL. Examples of such tasks include Question Answering(III-D1) and Co-reference Resolution(III-D2). The second category evaluates the model's proficiency in rephrasing sentences according to the provided instructions. Chat Completion(III-D3) serves as an instance of such tasks. In order to evaluate AXOLOTL, we use various metrics corresponding to each task. Following the suggestion by [23], we incorporate *toxicity* and *regard* scores as a metric to underscore the effectiveness of AXOLOTL on BOLD. For this purpose, we use a BERT-based model[3], that is trained on a large number of Wikipedia comments and offers toxicity scores for input text across all sensitive attributes.

[3] toxic-bert

According to [27], *regard*[4] aims to measure the sentiment directed towards a particular demographic group, rather than assessing the general sentiment of LM generated sentences. Their framework is designed specifically for sensitive attributes such as race, gender, and sexual orientation.

1) Question Answering: The objective of this task is to evaluate AXOLOTL at the discourse level through multiple-choice questions. After identifying bias (in form of an *orientation* to a demographic group and an *unpleasant characteristic*) and proposing a *pleasant resolution*, the model generates a new response with a lower bias. We utilize the Stereoset dataset, developed by [24], specifically designed for multi-choice question answering. Stereoset contains two types of sentences for each sensitive attribute: Intersentences and Intrasentences. For our task, we focus on Intersentences, where each data instance consists of a context sentence containing a target group and three corresponding sentences labeled as "stereotype", "anti-stereotype", and "meaningless". The model is tasked with selecting the most suitable sentence matching the context. We follow the bias detection, pleasant resolution identification, and self-debiasing steps outlined in Section II. Given the initial response r of the LLM model, the orientation to a group $\mathbf{g}_k$, the unpleasant characteristic (w^-), and the pleasant resolution (w^+), the model is prompted to identify a better response from the three options provided.

To evaluate the performance of the AXOLOTL on the Stereoset dataset, we introduce the Anti-Stereotype Score (*as-score). This metric is derived from the Stereotype Score proposed by* [24] *and is calculated as the ratio of stereotype responses to anti-stereotype responses* (as-score $= \frac{ss}{100-ss} \times 100$). The measures the tendency of the model to choose stereotypical options. A lower *as-score score suggests a preference for anti-stereotypical responses, indicating a reduction in bias. Ideally, a model with an* as-score score of 0 would demonstrate complete awareness of stereotypes and refrain from perpetuating biased scenarios. Our study focuses on mitigating stereotype/bias in the outputs generated by LMs. Therefore, we *as-score the effectiveness of* AXOLOTL *by measuring the reduction in* as-score after the rewrite.

Table II presents the *as-score results across all models and sensitive attributes before and after the rewrite. Our experimental findings reveal a visible decrease in* as-score across all models and attributes, signifying a decrease in stereotypical responses compared to anti-stereotype ones. Across all models and sensitive attributes, we observed that AXOLOTL effectively detected bias and provided substantial guidance in reducing the Anti-Stereotype Score (*as-score) by identifying stereotypical associations. Furthermore,* AXOLOTL *consistently outperformed* SELF-DIBIAS *in almost every comparison. Notably, in the profession attribute category,* AXOLOTL *facilitated significant decreases in* as-score: a reduction of 4.67 for GPT-3.5, 5.4 for Llama3-70b, and 2.93 for Llama3-8b. These results indicate that AXOLOTL was more effective compared to the increase observed with SELF-DIBIAS.

[4] Regard classifier

Sensitive Attributes	Model	Anti-Stereotype Score Before rewrite	Anti-Stereotype Score After rewrite	Anti-Stereotype Score Reduction
Gender	GPT-3.5	88.43	82.94	1.59
	Llama3-70b	1.07	1.00	0.07
	Llama3-8b	1.27	1.09	0.18
	Llama3-8b-SELF-DIBIAS	1.27	1.14	0.14
Race	GPT-3.5	70.57	66.97	3.6
	Llama3-70b	78.89	70.99	7.9
	Llama3-8b	95.42	90.36	5.06
	Llama3-8b-SELF-DIBIAS	95.42	87.72	7.7
Profession	GPT-3.5	99.60	94.93	4.67
	Llama3-70b	93.79	88.39	5.4
	Llama3-8b	95.38	92.45	2.93
	Llama3-8b-SELF-DIBIAS	95.38	98.25	-2.87

TABLE II: Results obtained from experiments conducted on the Stereoset dataset in Question Answering task.

	Multi-choice			**Open-ended**		
Group	**Male**	**Female**	**Neutral**	**Male**	**Female**	**Neutral**
GPT-3.5	0.359	0.105	0.536	0.283	0.196	0.521
GPT-3.5-AXOLOTL	0.074	0.101	0.825	0.118	0.109	0.773
llama3-70B	0.438	0.049	0.513	0.317	0.396	0.287
llama3-70B-AXOLOTL	0.031	0.0680	0.901	0.168	0.184	0.648
llama3-8B	0.258	0.300	0.442	0.327	0.234	0.439
llama3-8B-AXOLOTL	0.080	0.115	0.805	0.190	0.204	0.606
llama3-8B-SELF-DIBIAS	0.364	0.282	0.354	0.200	0.424	0.370

TABLE III: Portions of the answers with male and female pronouns on WinoBias dataset. We conduct experiments on both open-ended and multi-choice pronounce completion setup.

Group	**Race**		**Gender**		**Profession**	
regard	**Positive**	**Negative**	**Positive**	**Negative**	**Positive**	**Negative**
GPT-3.5	0.521	0.021	0.473	0.010	0.117	0.011
GPT-3.5-AXOLOTL	0.553	0.013	0.492	0.009	0.132	0.011
llama3-70B	0.353	0.021	0.388	0.013	0.134	0.012
llama3-70B-AXOLOTL	0.422	0.015	0.428	0.007	0.159	0.009
llama3-8B	0.493	0.030	0.236	0.017	0.086	0.012
llama3-8B-AXOLOTL	0.513	0.012	0.276	0.010	0.102	0.011
llama3-8B-SELF-DIBIAS	0.511	0.009	0.237	0.011	0.075	0.012

TABLE IV: Portions of texts classified as having positive and negative sentiment on BOLD dataset for the Chat Completion task.

Group	**Race**		**Gender**	
regard	**Positive**	**Negative**	**Positive**	**Negative**
GPT-3.5	0.892	0.058	0.848	0.045
GPT-3.5-AXOLOTL	0.899	0.055	0.856	0.042
llama3-70B	0.893	0.059	0.847	0.075
llama3-70B-AXOLOTL	0.905	0.053	0.855	0.060
llama3-8B	0.756	0.022	0.776	0.067
llama3-8B-AXOLOTL	0.761	0.016	0.786	0.059
llama3-8B-SELF-DIBIAS	0.775	0.031	0.790	0.052

TABLE V: Portions of texts classified as having positive and negative regard on BOLD dataset.

2) Co-reference Resolution: We structured the co-reference resolution experiment similarly to question answering, aiming to assess the capacity of the model to enhance its response from a provided set of options. The WinoBias dataset, created by Zhao et al. ([25]), is tailored to study gender bias within professions through a co-reference resolution system. Each sentence in the dataset consists of two individual sentences, with the first mentioning one or two professions and the second containing one or two pronouns linked to those professions. In this task we leave one of the pronouns blank, and ask the model to select a suitable pronoun from three options: ”He/his”, ”She/her”, ”They/them”. We refer to this task as Multi-choice Pronouns. Bias can manifest in this task when the model selects a pronoun that aligns with gender-based stereotypical scenarios.

For instance, the sentence ”[The lawyer] yelled at the hairdresser because [he] was mad.” demonstrates a common stereotype linking ”lawyer” with the male gender. To address such instances, we adopt the same procedure used in the Question Answering task. We provide the model with an instruction containing both w^- and w^+, guiding it to produce a more appropriate response. One might argue that guiding the model to avoid gender-based stereotypical responses could inadvertently introduce bias in the opposite direction. How-

Group	Race	Gender	Profession
GPT-3.5	2.14%	7.38%	3.71%
llama3-70B	8.48%	8.52%	2.17%
llama3-8B-AXOLOTL	6.01%	3.47%	3.20%
llama3-8B-SELF-DIBIAS	3.76%	4.24%	3.12%

TABLE VI: Toxicity score reduction percentage with respect to the original output on BOLD dataset.

ever, our approach in co-reference resolution not only aims to circumvent stereotypical scenarios but also strives to generate gender-neutral responses.

We also perform experiments on the WinoBias dataset using open-ended choice questions. In this task, we do not provide the model with pronouns as options, allowing it the freedom to complete the sentence with what it deems appropriate. This approach is particularly crucial for evaluating the model's performance without the constraint of predefined choices. We refer to this task as open-ended. Table III presents the results on the WinoBias dataset across all four models. These results indicate a notable decrease in gender-bias after the rewrite, with a higher percentage of our generated responses being gender-neutral. For instance, the results from llama3-70B in the multi-choice setup show a transition from 43.8% male and 4.9% female to 3.0% male, 6.8% female, and 90.1% neutral responses after revisions . This highlights the capability of our model to promote gender neutrality. A similar trend is observed in the open-ended results. Moreover, significant improvements were seen in smaller models like llama3-8B, which reached 80.5% gender neutralization after the rewrite, surpassing larger models such as llama3-70B, which only achieved 51.3% gender neutralization pre-rewrite in multi-choice setting. Furthermore, our model proved to be more effective in achieving neutralization compared to SELF-DIBIAS. In both the open-ended and multi-choice formats, as shown in Table III, SELF-DIBIAS actually led to an increase in the proportion of gendered responses and a decrease in neutral ones, thereby increasing the risk of stereotypical answers.

a) Impact of limited choices on Co-reference Resolution: In our experiments, chat completion is treated as an open-ended task, whereas question answering and co-reference resolution are structured as multi-choice tasks. Therefore, we chose to focus on co-reference resolution to explore the effects of removing predefined options, allowing the model the freedom to complete the task based on its own judgment. Table III presents the outcomes in an open-ended setting. The pattern of reduction in gendered responses and the trend toward neutralization mirror those observed in the original multi-choice version of co-reference resolution, where the model is prompted to choose among different pronouns. However, we noted a decrease in the extent of this increase. For example, the llama3-8B model produced nearly 40% neutral responses. In a multi-choice setting, the AXOLOTL increased the rate of neutral responses to approximately 80%, which is almost double that of the pre-rewrite. In contrast, in the open-ended setting, the percentage of neutral responses rose to 60%. Based on this pattern across all models, it is reasonable to conclude that limiting the range of choices assists AXOLOTL in regenerating unbiased responses.

3) Chat Completion: The second set of tasks aimed to evaluate AXOLOTL's ability in conversational setting and generating coherent responses. Given the debiasing instruction the model should be able to maintain the context of a conversation. These instructions include identifying the unpleasant characteristic (w^-) and suggesting the pleasant resolution (w^+) for the model to integrate during the rewrite phase.

In the Chat Completion task, each prompt from the dataset requires the model to complete the text, essentially making each dataset instance a "prefix" for a paragraph. The BOLD dataset, contains sentences ranging from 6 to 9 words across various domains from Wikipedia. We focus on domains related to race, gender, and profession.

a) Evaluation metrics.: As recommended by ([23]), we use *sentiment*, *toxicity*, and *regard* as our evaluation metrics. Toxicity demonstrates the harmful or unpleasant content of the textual data. The *toxicity* classifier labels textual data using a numerical value between 0 and 100. The *regard* and *sentiment* classifiers produce outputs categorized as "positive", "negative", or "neutral". It is crucial to distinguish between *regard* and *sentiment*. Regard precisely captures the sentiment toward a demographic group, while *sentiment* represents the overall sentiment of the sentence. Hence, *regard* serves as a measure of bias ([27]) with a sentence marked as negative by the *regard* classifier indicating a tendency toward negative representation of a demographic group. This indicates the presence of harmful bias in the sentence. As our ultimate goal is to mitigate the harmful bias produced by the model, we prioritize reducing the proportion of the results generated by AXOLOTL labeled as negative by the regard classifier.

b) Regard analysis.: Table V presents the experiment results across four models and three sensitive attributes in BOLD. It is evident that following our method, negative *regard* has decreased in nearly all instances, with minimal changes observed in positive *regard*. Notably, for the gender attribute, this reduction is as substantial with (0.015) decrease compared with as the original regard score , in the results produced by Llama3-70B. This means that 20% of the textual data that was labeled as negative before rewrite, was detected positive by the *regard* classifier post-rewrite. It should be highlighted that AXOLOTL managed to either maintain or enhance positive *regard* while reducing negative sentiments towards demographic groups. This experiment verifies that AXOLOTL successfully achieved its goal with decreasing the harmful bias towards protected groups.

c) Sentiment analysis.: In contrast to the regard analysis, our attention here is directed towards the positive portion of the model-generated responses. As previously discussed, *sentiment* signifies the overall polarity of the sentence, indicating whether it leans towards positive or negative. Thus, a sentence labeled as positive conveys a positive message. Given that we have reduced harmful bias through the regard analysis, a higher percentage of positive *sentiment* suggests an improvement in the responses generated by AXOLOTL.

Table IV showcases the results obtained from the *sentiment* classifier across all models and sensitive attributes. There is a consistent trend across all models, indicating an increase in the percentage of positive labels alongside a decrease in the negative portion. Additionally, our method turned out effective in boosting the performance of smaller models. This enhancement is clearly noticeable in the results of llama3-8B. For example, in the BOLD-race task, before applying our method, llama3-8B displayed the highest percentage of negative sentiment among the models tested, including llama3-70B and GPT-3.5. Following the application of our method, llama3-8B exhibited a noticeable reduction in negative sentiment responses compared to other models. Our approach improved the outputs by increasing the positive sentiment and reducing the negative, unlike SELF-DIBIAS, which failed to achieve similar improvements, especially in the BOLD-profession task.

d) Toxicity Analysis.: The toxicity classifier evaluates content for unpleasant, harmful, or disrespectful elements and assigns a score between 0 and 100 to each sentence. Therefore, a decrease in toxicity indicates a superior performance of the model. Table VI displays the percentage reduction in toxicity for each model post-rewrite with AXOLOTL compared to the pre-rewrite version across various sensitive attributes. While reductions were observed across all models and sensitive attributes, llama3-8B exhibited higher success rate in detecting and mitigating toxicity using our method in compare with SELF-DIBIAS. For instance, for the race attribute, llama3-8B-AXOLOTL reduced toxicity by 6% while the results from llama3-8B-SELF-DIBIAS show 3.76% improvement post-rewrite. Overall, our results demonstrate that our method was particularly effective in identifying toxicity within BOLD-gender, with a maximum reduction of 8.52% in results generated by llama3-70B and 7.38% by GPT-3.5. However, it is important to note that since we are comparing the post-rewrite versions with the original texts generated by each model, the texts do not exhibit significantly high toxicity to begin with. That is due to the internal settings designed within every model to prevent toxic behavior. This explains why the percentage improvements in many cases are relatively small.

IV. RELATED WORK

Research into human-like bias in Large Language Models is an ongoing endeavor aimed at addressing bias-related challenges from multiple perspectives. Bias can infiltrate LLMs through various channels, including data annotation via crowdsourcing (28; 29; 30), dataset diversity across demographic groups (4; 31), and selecting models that amplify specific parts of the dataset, potentially overlooking certain demographic groups (e.g., models tailored for English-speaking users) (32; 33). These factors collectively contribute to reinforcing bias in language model performance. Bias can manifest in embeddings and can be addressed at both the word-level and sentence-level. A significant body of research focuses on addressing and mitigating existing bias in word-level (8; 34; 35; 36) and sentence-level representations (6; 37; 38). To address bias, researchers have proposed various methods that tackles the bias in different phases of pre-training, in-training and post-training. We briefly review some examples of the state of the art methods.

a) Pre-training: Counterfactual Data Augmentation (CDA) (3) and data augmentation using demographic perturbation (39) aim to diminish bias by balancing training datasets. These methods require additional training on supplemental data points. Gehman et al. (40) proposes a domain-adaptive approach that pre-trains the model on a non-toxic dataset to avoid generating biased outputs. Both data augmentation and pre-training language models can be costly (10).

b) In-training: Regularization techniques like dropout (41) are used to further reduce gendered correlations during training. Regularization can also involve tuning the loss term to decrease the distance between the embeddings of a gender or race word and its equivalent in the loss function (42). This approach is particularly suitable when there is no access to training data but can also be used alongside CDA. Some in-training approaches are designed to debias pre-trained models for efficient fine-tuning while preserving the model's distributional knowledge. These methods include modifying the architecture and incorporating modules with a lower number of parameters into the layers of pre-trained language models (PLMs) (43).

c) Post-training: Schick et al.(26) demonstrates that language models are aware of their biases and can recognize undesired behaviors, such as biases or stereotypes, when provided with a description. Consequently, they can self-diagnose and then self-debias by reducing the probability of generating undesirable tokens. Despite this, studies have indicated that:

- Methods that require additional training are often costly.
- Many existing methods compromise the quality of the generated language model response (10; 44).
- Several existing methods are constrained to particular tasks (45) or specific sensitive attributes (10).
- Nearly all current research relies on open-source models, necessitating access to the models' internal configurations (9; 26).

Ebrahimi et al.(46) proposes REQUAL-LM, a post-training Monte Carlo-based method that treats LLMs as a black box. However, REQUA-LM does not attempt to debias the output but focuses solely on selecting the most fair and reliable response from a sample of responses. Our method is inspired by query rewriting techniques for prompt rewriting (12; 13; 14; 15; 16), as well as zero-shot learning techniques that leverage task descriptions (18). To the best of our knowledge, the closest work to ours is by (26). Schick et al. (26) defines the self-debiasing process as using the model's internal knowledge to reduce the probability of generating biased texts. This approach is feasible only with open-source language models where we have access to token probabilities. *Our method stands out as the first of its kind*, as it *does not require pre-training, fine-tuning, or accessing internal configurations (e.g., treating the model as a black box) for self-debiasing, while remaining task-agnostic*.

V. Conclusion

In this study, we introduced AXOLOTL, a novel post-processing framework designed to mitigate biases in Large Language Model (LLM) outputs. By leveraging self-debiasing techniques, AXOLOTL operates as a task-agnostic and model-agnostic tool and addresses key challenges in bias mitigation without compromising computational efficiency or model performance. Through a three-step process resembling zero-shot learning, AXOLOTL effectively identifies and corrects biases in LLM outputs, ensuring fairer outcomes across various applications. By treating LLMs as "black boxes" and utilizing public APIs, AXOLOTL offers broader applicability and ease of use, making it a valuable tool for practitioners seeking to address bias in natural language processing systems. Future research can further explore the scalability and generalizability of AXOLOTL across different LLM architectures and applications, ultimately advancing the goal of creating more equitable and inclusive AI systems.

VI. Limitations

In recognizing the limitations of our study, it is crucial to understand that the success of our approach closely depends on the effectiveness of embedding vectors ([22]) and their ability to capture and reflect subtle semantic biases in language. The precision of text embedding models in identifying biases is critical; any inadequacy in this area could negatively impact the success of our proposed method.

Furthermore, the integrity and selection of word sets (T^{+}, T^{-}) are crucial for the model's success in identifying biases and suggestion viable resolutions. Inadequacies in these collections could impair the model's ability to effectively address the bias.

Although AXOLOTL introduces a robust mechanism for mitigating bias, it does not assure absolute eradication of bias. It serves as a post-processing technique that operates without altering the foundational parameters of the underlying model, thereby not addressing the model's inherent biases directly.

Moreover, the implementation of AXOLOTL as an online framework necessitates network access to interact with Language Models via public APIs. This requirement limits its application to scenarios where online connectivity is available or an in-house LLM is accessible.

*Ethical Statement

This work fully complies with the ACL Ethics Policy. To the best of our knowledge, there are no ethical issues in this paper. We do not claim that we can entirely resolve the problem of bias in Language Models. Instead, we offer a framework that detect bias orientation and unpleasant characteristic in an LLM output, suggests a pleasant resolution, and applies self-debiasing.

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